A THREE-STAGE REAL-TIME FACE ANTI-SPOOFING SYSTEM

A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree

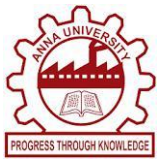
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BONAFIDE CERTIFICATE

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ABSTRACT

Face presentation attacks such as printed photographs, replayed videos, and digital screen displays pose a significant threat to modern biometric authentication systems. Traditional single-stage anti-spoofing methods often struggle to balance detection accuracy with real-time performance, especially under diverse attack conditions. To address these challenges, this work proposes a multi-stage deep learning pipeline for real-time face anti-spoofing detection that integrates lightweight screening, forensic-level analysis, and temporal motion verification within a unified framework.

The processes video streams at 30 frames per second and employs a cascaded architecture consisting of three sequential decision stages. The first stage utilizes a MobileNetV3-based fast screening model with a Laplacian variance pre-filter to perform rapid single-frame liveness detection. Ambiguous cases are escalated to the second stage, where a dual-branch forensic analysis network (ForensicNet) combines spatial texture extraction using EfficientNet with frequency domain analysis through a learnable Fast Fourier Transform (FFT) convolutional branch to detect Moire patterns .

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LIST OF ABBREVATION

AI	ARTIFICIAL INTELLIGENCE
BiLSTM	BIDIRECTIONAL LONG SHORT-TERM MEMORY
CNNs	CONVOLUTIONAL NEURAL NETWORKS
SVM	SUPPORT VECTOR MACHINE
DL	DEEP LEARNING
DNN	DEEP NEURAL NETWORK
HTER	HALF TOTAL ERROR RATE
F1	F1 SCORE
FL	FOCAL LOSS
STFM	SPOOFING STYLE FUSION MODULE
GMP	GLOBAL MAX POOLING
LSTM	LONG SHORT-TERM MEMORY
MSE	MEAN SQUARED ERROR